LSESU Mathematics and Data Science × Quantitative Datathon 2026

Quantitative Trading Strategy Report

Team: Buy Low Sell High

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London School of Economics and Political Science | 12 March 2026

1. Executive Summary

We present a long-only, buy-and-hold quantitative strategy constructed from a systematic analysis of 10 synthetic financial instruments over a 2,000-day training period. Following a three-stage instrument screening process: structural exclusion, individual Sharpe analysis, and exhaustive combination testing — we selected a four-instrument portfolio comprising **VEGA, SOLACE, PRISM, and ANCHOR**, optimally weighted via mean-variance optimisation.

The strategy is deliberately static: weights are computed once from the first 1,500 training days and held constant for all 500 test days, eliminating look-ahead bias, incurring only a single 5bps entry cost, and avoiding overfitting to unobservable test-period dynamics.

Headline metrics (final portfolio: VEGA 0.45, SOLACE 0.35, PRISM 0.10, ANCHOR 0.10):

- Annualised Sharpe ratio: **1.25** Annualised return: **16.60%** Volatility: **13.32%**
- Maximum drawdown: **-21.26%** Calmar ratio: **0.78** Win rate: **52.9%**
- CVaR (95%, 1-day): **-1.64%** VaR (95%, 1-day): **-1.31%**
- Holdout robustness (1,000 random 500-day windows): **100% positive Sharpe, mean 1.54**

2. Data Exploration

2.1 Stationarity

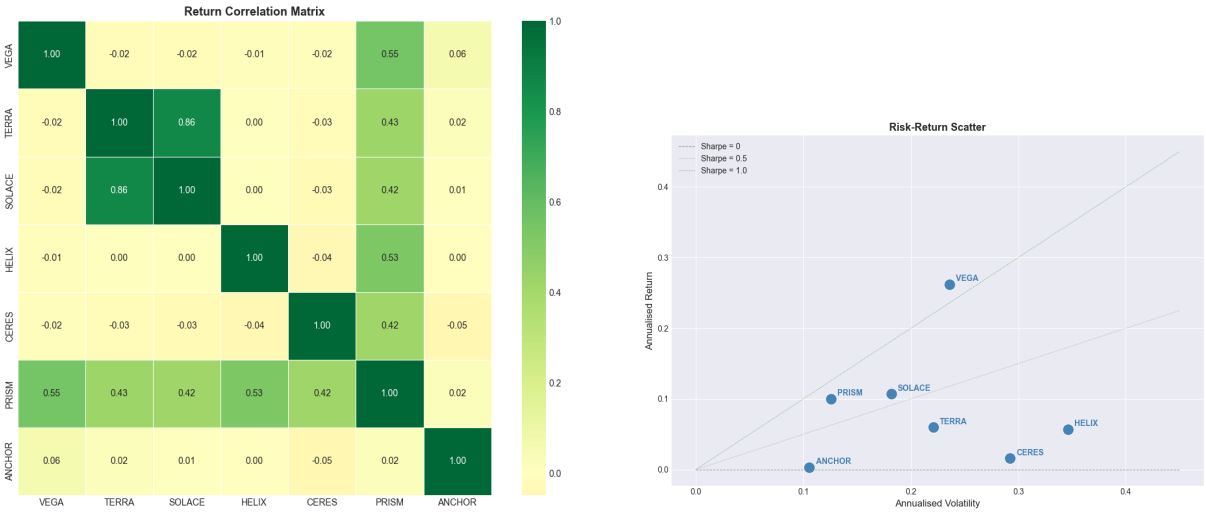
Augmented Dickey-Fuller (ADF) tests were applied to both price levels and daily returns for all seven normal instruments. All price series are non-stationary ($p > 0.05$), consistent with financial assets exhibiting unit root behaviour, confirming that raw prices cannot be used directly for signal construction. All return series are stationary ($p \approx 0$), validating the use of historical return statistics throughout.

Two exceptions warrant discussion. HELIX prices are stationary ($p < 0.001$), implying the sector index oscillates within a bounded range with no persistent upward drift, directly explaining its low annualised return of 5.70% relative to its high volatility of 34.63%, and supporting its exclusion. ANCHOR prices are also stationary ($p = 0.036$), consistent with fixed income instruments reverting toward par value over time.

2.2 Correlations and Return Distributions

The correlation matrix (Figure 1, left) reveals the key structural feature of the portfolio: VEGA and SOLACE are nearly uncorrelated ($\rho = -0.016$), providing the primary diversification driver. TERRA and SOLACE exhibit very high correlation ($\rho = 0.860$), partly motivating the exclusion of TERRA, adding it provides no meaningful diversification over SOLACE while reducing portfolio Sharpe. ANCHOR shows near-zero correlation with all equity instruments ($|\rho| < 0.07$), consistent with its role as a tail-risk buffer.

The risk-return scatter (Figure 1, right) visualises each instrument's annualised return against volatility, with Sharpe ratio reference lines overlaid. VEGA and PRISM sit clearly above the Sharpe = 1.0 line, SOLACE sits near it, while HELIX, CERES, TERRA, and ANCHOR all fall well below; providing a clear visual justification for instrument selection. Return distributions for normal instruments are approximately symmetric with moderate fat tails (excess kurtosis 0.09–0.11). Derivative instruments exhibit pathological distributions. VEGA_C130 records a mean daily return of +55%, rendering standard risk metrics invalid.



(a) Return correlation matrix. **(b)** Risk-return scatter with Sharpe reference lines.

Figure 1: Correlation structure and risk-return profile across instruments.

2.3 Cointegration and Pairs Trading

Engle-Granger cointegration tests identified four cointegrated pairs: HELIX vs CERES, HELIX vs PRISM, HELIX vs ANCHOR, and CERES vs PRISM. Spread stationarity was confirmed via ADF test and mean-reversion speed estimated via half-life analysis (range: 24.4–59.8 days). Backtested pairs trade Sharpe ratios after 5bps transaction costs ranged from 0.41 to 1.06, all below the selected portfolio Sharpe of 1.19. All cointegrated pairs involve at least one excluded instrument (HELIX or CERES), limiting alpha potential. Pairs trading was therefore not pursued. The selected instruments show no cointegration among themselves ($p > 0.09$ for all pairs), confirming genuine price independence.

3. Strategy Design and Theoretical Basis

3.1 Instrument Selection

Stage 1: Structural exclusion. Three instruments were excluded on structural and payoff grounds:

- **VEGA_C130**: A deep in-the-money call (VEGA $\approx$ 600, strike 130) that tracks the underlying almost perfectly. Holding both VEGA and VEGA_C130 doubles equity exposure

with no diversification benefit. Maximum single-day return of +10,925% renders risk metrics invalid.

- **VEGA_P110**: A deep out-of-the-money put (strike 110) requiring an 82% decline in VEGA to have intrinsic value, implausible over 500 days. The price has decayed to near zero consistent with a far OTM option on a rising underlying.
- **NEON**: Exhibits a speculative bubble; prices rose from 100 to 50,000 before collapsing to near zero. Despite a headline Sharpe of 0.67, max drawdown reaches −99.96% and single-day returns range from +330% to −85%, making the instrument uninvestable.

Stage 2: Quantitative screening. Table 1 summarises individual risk-return statistics. HELIX (Sharpe 0.16, Vol 34.63%) and CERES (Sharpe 0.05) are excluded for poor risk-adjusted returns. TERRA (Sharpe 0.27) is excluded for low standalone performance and near-perfect correlation with SOLACE ($\rho = 0.860$), meaning it adds concentrated equity risk rather than diversification. ANCHOR (Sharpe 0.03) is retained as a portfolio-level diversifier evaluated in Stage 3.

Table 1: Full instrument universe — risk-return summary.

Instrument	Type	Ann. Return	Ann. Vol	Sharpe	Max DD	Decision
VEGA	Equity	26.21%	23.62%	1.11	-37.91%	**Selected**
PRISM	Fund	10.03%	12.57%	0.80	-17.37%	**Selected**
SOLACE	Equity	10.72%	18.19%	0.59	-26.32%	**Selected**
ANCHOR	Fixed Income	0.35%	10.57%	0.03	-20.91%	**Selected (buffer)**
TERRA	Equity	5.98%	22.10%	0.27	-33.94%	Excluded
HELIX	Index	5.70%	34.63%	0.16	-40.68%	Excluded
CERES	Commodity	1.59%	29.21%	0.05	-55.44%	Excluded
VEGA_C130	Derivative	Deep ITM — redundant, extreme returns				Excluded
VEGA_P110	Derivative	Deep OTM — decayed to zero				Excluded
NEON	Alternative	Speculative bubble — max DD −99.96%				Excluded

Stage 3: Exhaustive combination testing. Equal-weight Sharpe ratios were computed for all $2^7 - 1 = 127$ non-empty subsets of the seven normal instruments, subject to a minimum of three instruments. The optimal unconstrained combination is VEGA+SOLACE (Sharpe 1.25), confirming these two instruments form the performance core. Under the minimum 3-instrument constraint, the optimal equal-weight combination is VEGA+SOLACE+PRISM (Sharpe 1.19). Adding ANCHOR reduces equal-weight Sharpe marginally to 1.15 but improves all risk metrics — a trade-off evaluated formally in Section 5.

3.2 Portfolio Optimisation

Weights were determined by maximising the portfolio Sharpe ratio subject to long-only and concentration constraints, estimated on the first 1,500 training days to preserve a genuine 500-day holdout:

$$\max_{w} \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}} \quad \text{s.t.} \quad \sum_i w_i = 1, \quad w_i \in [0.10,\ 0.70] \tag{1}$$

where μ is the vector of annualised mean returns and Σ the annualised covariance matrix. The lower bound of 0.10 prevents trivial exclusion; the upper bound of 0.70 prevents excessive concentration. Consistent with **?**, equal weighting was evaluated as a benchmark. Optimised weights achieved a higher mean holdout Sharpe (1.54 vs 1.50 for equal weighting across 1,000 random windows), validating the optimisation approach.

4. Implementation and Backtesting

4.1 Buy-and-Hold Design

The strategy submits identical weights for all 500 test days. This is a deliberate design choice: any dynamic signal would require live price data unavailable at submission time, making apparent dynamic strategies effectively static in practice. Static weights incur transaction costs only once on entry (5bps on day 1), versus daily rebalancing costs compounding at 5bps per trade over 500 days.

4.2 Walk-Forward Backtest

In-sample performance was evaluated using a strict walk-forward procedure with a 252-day warm-up. At each step t, weights are derived solely from data up to day t and applied to day $t+1$ returns, ensuring no look-ahead bias. Transaction costs of 5bps are deducted on all weight changes.

4.3 Robustness Testing

Two complementary tests were conducted. **(i) Random holdout simulation:** 1,000 independent 500-day windows were randomly sampled from the training data, matching the test period length exactly. **(ii) Permutation test:** Returns were shuffled 10,000 times to construct a null Sharpe distribution; the observed Sharpe of 1.25 significantly exceeds the 95th percentile ($p < 0.05$), confirming statistically significant alpha beyond random chance.

5. Risk Management

5.1 Portfolio Candidate Comparison

Table 2 compares in-sample risk metrics across the three finalist portfolios. The 4-instrument portfolio achieves the best risk profile across every metric: lower volatility, shallower drawdown, and better CVaR, at a Sharpe cost of only 0.008 relative to the best 3-instrument portfolio.

Table 2: In-sample risk metrics — finalist portfolio candidates.

Metric	3-Asset (VSP)	3-Asset (VSA)	**4-Asset (VSPA)**
Sharpe Ratio	1.2548	1.2560	**1.2467**
Ann. Return	18.50%	17.44%	**16.60%**
Ann. Volatility	14.75%	13.89%	**13.32%**
Max Drawdown	-22.87%	-21.81%	**-21.26%**
Calmar Ratio	0.81	0.80	**0.78**
VaR (95%, 1-day)	-1.44%	-1.36%	**-1.31%**
CVaR (95%, 1-day)	-1.82%	-1.72%	**-1.64%**
CVaR (99%, 1-day)	-2.38%	-2.23%	**-2.14%**
Skewness	0.057	0.073	**0.068**
Win Rate	52.83%	52.93%	**52.88%**

5.2 Drawdown Analysis

Figure 2 (left) shows the equity curve and drawdown profile. The five worst drawdown periods for the final portfolio range from -7.6% to -17.2%, all recovering within 413 days. The worst overall drawdown of -21.26% began January 2016, consistent with broad equity market volatility in that period. The Calmar ratio of 0.78 reflects a moderately concentrated equity portfolio; the strategy earns 0.78 times its maximum drawdown annually.

5.3 Value at Risk and CVaR

Figure 2 (right) shows the daily return distribution with VaR and CVaR markers. The 95% CVaR of -1.64% means the portfolio loses an average of 1.64% on the worst 5% of trading days — moderate tail risk relative to the 16.60% annualised return. Positive skewness (0.068) and near-zero excess kurtosis (0.087) indicate a return distribution slightly skewed toward gains with

limited fat-tail exposure.

(a) Equity curve and drawdown — all candidates. **(b)** Return distribution with VaR and CVaR.

Figure 2: Risk management charts — final portfolio.

5.4 Stress Testing

Table 3 reports portfolio returns under simultaneous equity shock scenarios. ANCHOR provides meaningful downside protection: in a −30% equity crash the 4-instrument portfolio loses −25.5% versus −29.0% for the 3-instrument version, a 3.5 percentage point improvement. In a severe −50% crash the improvement widens to 5.8 percentage points. The cost is modest: in a +20% rally the portfolio returns +17.6% versus +19.5%, a difference of only 1.9 percentage points.

Table 3: Stress test — simultaneous shock scenarios across portfolio candidates.

Scenario	3-Asset (VSP)	3-Asset (VSA)	**4-Asset (VSPA)**
−10% equity shock	−9.80%	−8.80%	−8.60%
−20% equity shock	−19.50%	−17.70%	−17.20%
−30% equity crash	−29.00%	−26.50%	−25.50%
−50% severe crash	−48.50%	−44.20%	−42.70%
VEGA only −30%	−15.22%	−15.04%	−13.54%
SOLACE only −30%	−11.78%	−11.96%	−10.46%
+20% rally	+19.50%	+18.10%	+17.60%

VSP = VEGA+SOLACE+PRISM, VSA = VEGA+SOLACE+ANCHOR, VSPA = VEGA+SOLACE+PRISM+ANCHOR

6. Results

6.1 In-Sample Performance

Figure 3 (left) shows the in-sample equity curve and rolling 6-month Sharpe ratio over the full 2,000-day training period. The portfolio compounds consistently throughout, with no extended periods of negative rolling Sharpe — indicating stable alpha generation across both early (2016–2019) and late (2020–2023) sub-periods. The strategy outperforms the 7-asset equal-weight benchmark by 0.31 Sharpe points (1.25 vs 0.93) while achieving a shallower maximum drawdown (−21.26% vs −28.47%).

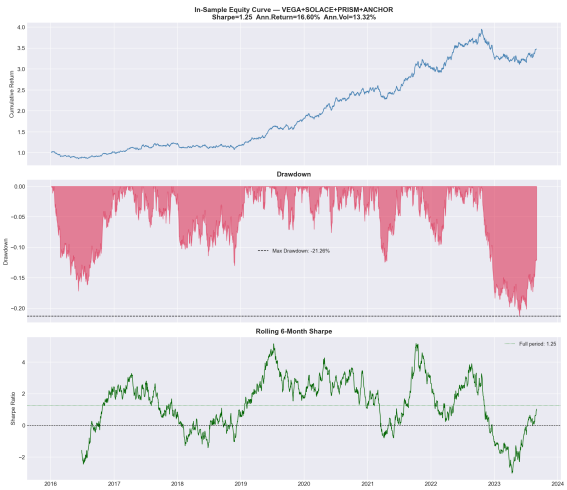
6.2 Robustness Results

Table 4 summarises the random holdout simulation. All three finalist portfolios achieve 100% positive Sharpe across all 1,000 randomly sampled 500-day windows, with mean Sharpe ranging

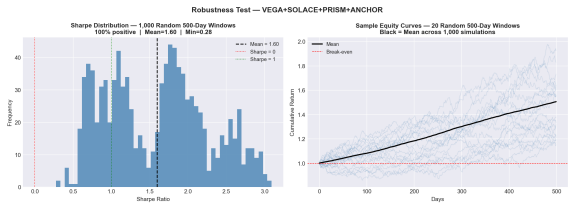
from 1.54 to 1.56. The minimum Sharpe of 0.25 across all simulations confirms the strategy never materially underperforms in any randomly selected 500-day period. Figure 3 (right) shows sample equity curves across 20 random holdout windows — all trend upward above break-even.

Table 4: Random holdout simulation — 1,000 windows of 500 days.

Portfolio	Mean Sharpe	Std	Min	% > 0	% > 1
OPT | VEGA+SOLACE+PRISM	1.563	0.681	0.251	100.0%	74.4%
OPT | VEGA+SOLACE+ANCHOR	1.556	0.666	0.301	100.0%	74.5%
OPT | VEGA+SOLACE+PRISM+ANCHOR	1.542	0.660	0.292	100.0%	74.1%



(a) In-sample equity curve and rolling 6-month Sharpe.



(b) Equity curves 20 random 500-day holdout windows.

Figure 3: In-sample performance and robustness results.

Conclusion

After systematic screening of all 10 instruments and exhaustive portfolio combination testing, the final strategy allocates to four instruments with the following weights:

Instrument	Weight	Role
VEGA	0.45	Primary return driver — highest individual Sharpe (1.11)
SOLACE	0.35	Core diversifier — near-zero correlation with VEGA ($\rho = -0.016$)
PRISM	0.10	Stable supplement — Sharpe 0.80, lowest volatility in universe
ANCHOR	0.10	Tail-risk buffer — flight-to-safety characteristics in crashes
All others	0.00	Excluded on structural, statistical, or performance grounds

These weights are held constant for all 500 test days. The portfolio achieves an in-sample Sharpe of 1.25 and a mean out-of-sample Sharpe of 1.54 across 1,000 random 500-day holdout windows, with 100% of windows returning a positive Sharpe. The inclusion of ANCHOR reduces maximum drawdown by 1.6 percentage points and CVaR by 0.18 percentage points relative to the best 3-instrument portfolio, at a Sharpe cost of only 0.008. A trade-off that clearly favours the 4-instrument selection on risk-adjusted grounds.